

Solutions Week 4

1 Gates for universal classical computation

- (a) Every Boolean function has a Disjunctive Normal Form (DNF):

$$f(x_1, \dots, x_n) = \bigvee_{a \in f^{-1}(1)} \bigwedge_{i=1}^n \ell_i^{(a)},$$

where each literal $\ell_i^{(a)}$ is either x_i or $\neg x_i$ depending on whether the i th bit of a is 1 or 0.

Thus each conjunction uses AND and NOT, and the outer combination uses OR. Hence AND, OR, NOT suffice to compute any Boolean function.

- (b) Recall:

$$\text{NAND}(x, y) = \neg(x \wedge y).$$

Observe:

$$\neg x = \text{NAND}(x, x),$$

$$x \wedge y = \neg \text{NAND}(x, y),$$

$$x \vee y = \neg(\neg x \wedge \neg y).$$

Since NOT and AND can be built from NAND, and AND/OR/NOT are universal, NAND alone is universal.

Similarly, NOR satisfies:

$$\neg x = \text{NOR}(x, x),$$

$$x \vee y = \neg \text{NOR}(x, y),$$

$$x \wedge y = \neg(\neg x \vee \neg y).$$

Hence NOR alone is also universal.

¹Problems are taken from the homework sheets of Ryan O'Donnell's course in 2018; <https://www.cs.cmu.edu/~odonnell/quantum18/>

(c) Circuits using only AND and OR are monotone: if $x \leq y$ coordinatewise, then $f(x) \leq f(y)$.

Functions like XOR are not monotone. Since there are infinitely many non-monotone Boolean functions, infinitely many cannot be computed using only AND and OR.

(d) Circuits using XOR and NOT compute exactly affine functions:

$$f(x) = a_0 \oplus a_1 x_1 \oplus \cdots \oplus a_n x_n.$$

These are linear functions over $\mathbb{F}_2$. There are only 2^{n+1} such functions, but there are 2^{2^n} total Boolean functions. Hence infinitely many cannot be computed.

2 Computational arithmetic

We use repeated squaring.

Write C in binary:

$$C = \sum_{i=0}^k c_i 2^i.$$

Compute:

$$B^{2^0} \bmod D, \quad B^{2^1} \bmod D, \quad B^{2^2} \bmod D, \dots$$

Each step:

$$B^{2^{i+1}} \equiv (B^{2^i})^2 \pmod{D}.$$

Then

$$B^C \equiv \prod_{i:c_i=1} B^{2^i} \pmod{D}.$$

There are $O(n)$ squarings and multiplications and modulo operations. Each operation of n -bit numbers costs $O(n^2)$ steps (grade-school) and $\tilde{O}(n)$ using advanced methods.

Hence total time:

$$O(n \cdot n^2) = O(n^3).$$

Thus modular exponentiation is in P.

3 Simulating a biased coin

- (a) Any procedure using only $\text{FLIP}_{1/2}$ produces probabilities of 1 of the form $k/2^m$. This is because say you flip a $\text{FLIP}_{1/2}$ a finite n many times and via a deterministic algorithm you can map every string in $\{0,1\}^n$ to either 0 or 1. Since $1/3$ cannot equal $k/2^m$ for integers k, m , exact simulation is impossible.

- (b) Algorithm:

Repeat:

```
    Flip two fair coins to get number r in {0,1,2,3}
    If r=0: return 0
    If r=1: return 1
    If r=2: return 0
    If r=3: continue
```

Here:

$$\Pr[\text{FAIL per round}] = 1/4.$$

Conditioned on success ($r=0,1,2$), we have:

$$\Pr[1] = 1/3.$$

Repeat at most k times. Failure probability after k rounds is $(1/4)^k$. Choose k such that $(1/4)^k \leq \varepsilon$.

- (c) No, it does not if you are using the coin at most $\text{poly}(n)$ times where n is the input size. You can truncate the real number p (in its binary representation) after n slots to obtain a new bias p' . Note, a coin with bias p' can be exactly simulated with a $\text{FLIP}_{1/2}$ coin and that introduces a maximum error of $\frac{1}{2^n}$. If we repeat it $\text{poly}(n)$ times, the maximum error is $\frac{\text{poly}(n)}{2^n}$ which is still negligible.

4 Dealing with error

- (a) Repeat C independently k times. Output '?' only if all runs fail.

Failure probability becomes:

$$(0.9)^k.$$

Choose $k = 5000$ to get below 2^{-500} .

- (b) Repeat C until it does not output '?'.

Expected repetitions:

$$\frac{1}{1 - 0.9} = 10.$$

Hence expected running time increases by constant factor.

- (c) Repeat independently k times. Output 1 if any run outputs 1.

Failure probability:

$$(0.9)^k.$$

Choose k as before.

- (d) Repeat C independently k times. Take majority vote.

By Chernoff bound:

$$\Pr[\text{majority wrong}] \leq e^{-2\delta^2 k}.$$

Since error per trial is 0.4, bias is 0.1. Thus exponential decay in k .

Choose $k = O(5000)$ to achieve 2^{-500} .

5 Complex number exercises

Try yourself.

6 Dirac notation

$$|\phi\rangle = 3|0\rangle - 5i|1\rangle.$$

- (a)

$$\langle\phi|\phi\rangle = 9 + 25 = 34.$$

- (b) Normalize by dividing by $\sqrt{34}$.

- (c) Probabilities:

$$P(0) = 9/34, \quad P(1) = 25/34.$$

- (d) Compute overlaps:

$$\langle +|\phi\rangle = \frac{1}{\sqrt{2}}(3 - 5i).$$

Probability:

$$\frac{|3 - 5i|^2}{2 \cdot 34} = \frac{34}{68} = \frac{1}{2}.$$

Similarly for $-$. Hence both outcomes probability $1/2$.

- (e) Check orthonormality directly via inner products.

7 Projectors and reflections

(a) If $|\psi\rangle = |0\rangle$, $|\phi\rangle = |+\rangle$:

$$Q = |+\rangle\langle 0| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}.$$

Opposite case similar.

(c) Q maps $|\psi\rangle$ to $|\phi\rangle$, and any orthogonal vector to 0.

(d) P is orthogonal projection onto $\text{span}\{|\psi\rangle\}$.

(e) $1 - 2P$ reflects across orthogonal complement. One checks:

$$(1 - 2P)^2 = 1 - 4P + 4P^2 = 1 - 4P + 4P = 1.$$

Hence unitary.

(f) Note, $(|\phi\rangle\langle\psi|)^\dagger = |\psi\rangle\langle\phi|$ Check:

$$UU^\dagger = \sum_i |\phi_i\rangle\langle\phi_i| = I.$$

$$U^\dagger U = \sum_i |\psi_i\rangle\langle\psi_i| = I.$$

8 Unitary matrices

If $\|A\psi\| = \|\psi\|$ for all ψ , then

$$\langle A\psi, A\psi \rangle = \langle \psi, \psi \rangle.$$

Hence

$$\langle \psi, A^\dagger A \psi \rangle = \langle \psi, \psi \rangle \implies \langle \psi | A^\dagger A - I | \psi \rangle = 0 \quad \forall \psi$$

Try to show this implies $A^\dagger A = I$. [This looks trivial, still try to show it formally.] Hence A unitary.

9 Quantum Anti-Zeno Effect

Define intermediate states:

$$|\psi_k\rangle = \cos \frac{k\pi}{2N} |0\rangle + \sin \frac{k\pi}{2N} |1\rangle.$$

Measure in basis $\{|\psi_k\rangle, |\psi_k^\perp\rangle\}$ sequentially for $k = 1, \dots, N$.

At each step success probability $\cos^2(\pi/2N)$. Total success probability:

$$\left(\cos^2 \frac{\pi}{2N}\right)^N \rightarrow 1 \text{ as } N \rightarrow \infty.$$

Thus almost surely reach $|1\rangle$.

10 GCD

- (a) If $Q|A$ and $Q|B$, then $Q|(A - B)$. Conversely, if $Q|(A - B)$ and $Q|B$, then $Q|A$. Hence

$$\gcd(A, B) = \gcd(A - B, B).$$

- (b) Since repeated subtraction equals remainder, algorithm is correct by induction.
(c) Use induction.

11 Perfect Powers

- (a) Use binary search for square root:

```
low=0, high=A
while low<=high:
    mid=(low+high)//2
    if mid^2=A: return mid
    if mid^2<A: low=mid+1
    else high=mid-1
return NOT SQUARE
```

Each multiplication costs $M(n)$. Binary search takes $O(n)$ steps. Total $O(nM(n))$.

- (b) For $C = 2$ to n :

Binary search for $B = \lfloor A^{1/C} \rfloor$. Check if $B^C = A$.

Exponentiation takes polynomial time. Since $C \leq n$, total time polynomial in n .

Solutions Week 5

1 Tensor Product Practice

- (a) Let $A \in \mathbb{C}^{m \times n}$. By definition,

$$A = \sum_{i=1}^m \sum_{j=1}^n A_{ij} |i\rangle \langle j|.$$

Indeed,

$$\langle i| A |j\rangle = \langle i| \left(\sum_{k,\ell} A_{k\ell} |k\rangle \langle \ell| \right) |j\rangle = \sum_{k,\ell} A_{k\ell} \langle i| k\rangle \langle \ell| j\rangle = A_{ij}.$$

Hence the matrix entries satisfy

$$A_{ij} = \langle i| A |j\rangle.$$

(b) By definition of the tensor product,

$$(A \otimes B)(|j\rangle \otimes |\ell\rangle) = (A|j\rangle) \otimes (B|\ell\rangle).$$

Taking inner product with $\langle i| \otimes \langle k|$ gives

$$\langle ik|A \otimes B|j\ell\rangle = \langle i|A|j\rangle \cdot \langle k|B|\ell\rangle.$$

This identity uniquely determines the matrix entries of $A \otimes B$, and hence matches the standard Kronecker product definition.

(c) Compute matrix entries:

$$\langle ik|(A \otimes B) \circ (C \otimes D)|j\ell\rangle = \langle ik|A \otimes B|j\ell\rangle \langle ik|C \otimes D|j\ell\rangle.$$

Using part (b):

$$= \langle i|A|j\rangle \langle k|B|\ell\rangle \langle i|C|j\rangle \langle k|D|\ell\rangle.$$

Rearranging,

$$= (\langle i|A|j\rangle \langle i|C|j\rangle) (\langle k|B|\ell\rangle \langle k|D|\ell\rangle).$$

Thus

$$(A \otimes B) \circ (C \otimes D) = (A \circ C) \otimes (B \circ D).$$

(d) Assume A and B invertible. Then

$$(A \otimes B)(A^{-1} \otimes B^{-1}) = (AA^{-1}) \otimes (BB^{-1}) = I \otimes I = I.$$

Similarly for reversed order. Hence

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}.$$

(e) Using definition of adjoint:

$$\langle ik|(A \otimes B)^\dagger|j\ell\rangle = \overline{\langle j\ell|A \otimes B|ik\rangle}.$$

Using part (b):

$$= \overline{\langle j|A|i\rangle \langle \ell|B|k\rangle} = \langle i|A^\dagger|j\rangle \langle k|B^\dagger|\ell\rangle.$$

Hence

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger.$$

(f) We compute:

$$\langle u_i \otimes v_j | u_{i'} \otimes v_{j'} \rangle = \langle u_i | u_{i'} \rangle \langle v_j | v_{j'} \rangle.$$

Since both families are orthonormal,

$$= \delta_{ii'} \delta_{jj'}.$$

Thus the collection is orthonormal. Since there are de such vectors in a de -dimensional space, they form an orthonormal basis.

2 1 ebit + 1 qubit $\geq$ 2 bits

(a) Start with

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Alice applies $X^u Z^v$ to her qubit.

One checks:

$$\begin{aligned} I &: |\Phi^+\rangle \\ X &: \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle) = |\Psi^+\rangle \\ Z &: \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |\Phi^-\rangle \\ XZ &: \frac{1}{\sqrt{2}}(|10\rangle - |01\rangle) = |\Psi^-\rangle \end{aligned}$$

Thus the four messages map to the four orthonormal Bell states:

$$\{\Phi^+, \Phi^-, \Psi^+, \Psi^-\}.$$

Since these are mutually orthonormal, Bob can measure in the Bell basis and perfectly determine uv .

(b) Bob performs the standard Bell measurement:

- (a) Apply CNOT (Alice qubit control, Bob target).
- (b) Apply Hadamard to first qubit.
- (c) Measure both in computational basis.

This maps:

$$\Phi^+ \mapsto 00, \quad \Phi^- \mapsto 01, \quad \Psi^+ \mapsto 10, \quad \Psi^- \mapsto 11.$$

Hence Bob recovers uv with probability 1.

3 Indistinguishable States

(a) Let $\{|\psi\rangle, |\psi^\perp\rangle\}$ be orthonormal. There exists a unitary U such that

$$U|0\rangle = |\psi\rangle, \quad U|1\rangle = |\psi^\perp\rangle.$$

Then

$$(U \otimes U) \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|\psi\psi\rangle + |\psi^\perp\psi^\perp\rangle).$$

But Bell states are invariant under $U \otimes U$. Hence

$$\frac{1}{\sqrt{2}}(|\psi\psi\rangle + |\psi^\perp\psi^\perp\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

(b) Suppose $|v\rangle = c|u\rangle$ with $|c| = 1$.

For any unitary U ,

$$U|v\rangle = cU|u\rangle.$$

If the measurement basis is $\{|\psi\rangle, |\psi^\perp\rangle\}$ Measurement probabilities depend on $|\langle\psi|U|\psi\rangle|^2$. Since global phase cancels:

$$|c|^2 = 1,$$

all outcome probabilities are identical.

It also holds in the most general case where we can append Ancilla qubits, do a joint unitary and a final measurement.

(c) Scenario 1 density matrix (density matrix of qubit state $|\psi\rangle$ is $|\psi\rangle\langle\psi|$ and the measurement probability of $|\psi\rangle$ in the basis $\{|\phi\rangle, |\phi^\perp\rangle\}$ is $|\langle\psi, \phi\rangle|^2 = \text{Tr}[|\psi\rangle\langle\psi| |\phi\rangle\langle\phi|])$:

$$\rho_1 = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I}{2}.$$

Scenario 2:

$$\rho_2 = \frac{1}{2}(|+\rangle\langle +| + |-\rangle\langle -|).$$

But

$$|+\rangle\langle +| + |-\rangle\langle -| = I.$$

Hence

$$\rho_2 = \frac{I}{2}.$$

Thus $\rho_1 = \rho_2$. No physical procedure can distinguish the scenarios.

4 Elementary Number Theory

Will be discussed in a video tutorial soon. Questions won't be asked on this topic in Quiz 1.

5 Hadamard Transform I

Let $S \subseteq [n]$ and let $y \in \{0, 1\}^n$ be its indicator vector.

Applying H to qubits in S gives:

Each $i \in S$:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

Thus final state:

$$\frac{1}{2^{|S|/2}} \sum_{x: x_i=0 \text{ if } i \notin S} |x\rangle.$$

Equivalently,

$$\frac{1}{2^{|S|/2}} \sum_{z \in \{0,1\}^{|S|}} |z \cdot y\rangle,$$

i.e., uniform superposition over all bitstrings supported on S .

In words:

It is the uniform superposition over all strings whose 1's occur only inside S , with amplitude $2^{-|S|/2}$.

Solutions Week 6

1 Quantum circuit practice

The circuit acts on three qubits:

Top qubit:

$$\alpha_0 |0\rangle + \alpha_1 |1\rangle$$

Second qubit:

$$|0\rangle$$

Third qubit:

$$|0\rangle$$

The gates (from left to right) are:

1. Hadamard on qubit 2. 2. CNOT with qubit 2 as control and qubit 3 as target. 3. CNOT with qubit 1 as control and qubit 2 as target. 4. Hadamard on qubit 1.

(a) Final state

Initial state:

$$(\alpha_0 |0\rangle + \alpha_1 |1\rangle) |0\rangle |0\rangle.$$

After applying H to qubit 2:

$$|0\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle),$$

so the state becomes:

$$(\alpha_0 |0\rangle + \alpha_1 |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle$$

Explicitly:

$$= \frac{1}{\sqrt{2}} (\alpha_0 |000\rangle + \alpha_0 |010\rangle + \alpha_1 |100\rangle + \alpha_1 |110\rangle).$$

Apply CNOT (qubit 2 $\rightarrow$ qubit 3):

This flips qubit 3 whenever qubit 2 is 1:

$$\frac{1}{\sqrt{2}} (\alpha_0 |000\rangle + \alpha_0 |011\rangle + \alpha_1 |100\rangle + \alpha_1 |111\rangle).$$

Apply CNOT (qubit 1 $\rightarrow$ qubit 2):

Flip qubit 2 whenever qubit 1 is 1:

$$= \frac{1}{\sqrt{2}} (\alpha_0 |000\rangle + \alpha_0 |011\rangle + \alpha_1 |110\rangle + \alpha_1 |101\rangle).$$

Finally apply Hadamard to qubit 1:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

After simplification, the final state becomes:

$$\frac{1}{2} \left[|00\rangle (\alpha_0 |0\rangle + \alpha_1 |1\rangle) + |01\rangle (\alpha_0 |1\rangle + \alpha_1 |0\rangle) + |10\rangle (\alpha_0 |0\rangle - \alpha_1 |1\rangle) + |11\rangle (\alpha_0 |1\rangle - \alpha_1 |0\rangle) \right].$$

(b) Measuring the top two qubits

Each of the four measurement outcomes

$$00, 01, 10, 11$$

occurs with probability:

$$\frac{1}{4}.$$

(Each branch has squared norm $1/4$ since $|\alpha_0|^2 + |\alpha_1|^2 = 1$.)

The post-measurement state of the third qubit is:

Outcome	Third qubit state
00	$\alpha_0 0\rangle + \alpha_1 1\rangle$
01	$\alpha_0 1\rangle + \alpha_1 0\rangle = X \psi\rangle$
10	$\alpha_0 0\rangle - \alpha_1 1\rangle = Z \psi\rangle$
11	$\alpha_0 1\rangle - \alpha_1 0\rangle = XZ \psi\rangle$

where $|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle$.

Thus the third qubit contains the original state up to a Pauli correction, depending on the classical measurement outcome.

2 Fun with gates

(a) Reverse CNOT using Hadamards

Identity:

$$(H \otimes H) \text{CNOT}_{A \rightarrow B} (H \otimes H) = \text{CNOT}_{B \rightarrow A}.$$

Proof:

Note $\text{CNOT}_{A \rightarrow B} = |0\rangle\langle 0|_A \otimes I_B + |1\rangle\langle 1|_A \otimes X_B$.

$\text{CNOT}_{B \rightarrow A} = I_A \otimes |0\rangle\langle 0|_B + X_A \otimes |1\rangle\langle 1|_B$. Now Check correctness as shown in tutorial.

(b) SWAP from CNOTs

Standard decomposition:

$$\text{SWAP} = \text{CNOT}_{A \rightarrow B} \text{CNOT}_{B \rightarrow A} \text{CNOT}_{A \rightarrow B}.$$

Direct verification on basis states confirms this.

3 Implausible consequences of superstrong nonlocality

(a) Computing inner product

For each i , use one nonlocal box with inputs x_i, y_i . Outputs satisfy:

$$a_i + b_i = x_i y_i \pmod{2}.$$

Alice computes

$$A = \sum_i a_i \pmod{2}$$

and Bob computes

$$B = \sum_i b_i \pmod{2}.$$

Then

$$A + B = \sum_i x_i y_i = IP_2(x, y).$$

Alice sends one bit A to Bob. Bob computes $A + B$.

(b) Polynomial mod 2 representation

Since AND is multiplication mod 2 and NOT is $1 + x$, every Boolean circuit corresponds to a polynomial over $\mathbb{F}_2$.

Hence every Boolean function has a polynomial representation mod 2.

(c)

Expand polynomial mod 2. Each monomial factors as:

$$(\text{product of } x_i) \cdot (\text{product of } y_j).$$

Thus write:

$$f(x, y) = \sum_j A_j(x) B_j(y).$$

(d)

For each term $A_j(x)B_j(y)$:

Use nonlocal box to compute product. Sum all contributions using parity trick as in (a).

Alice sends one bit containing total of her contributions. Bob reconstructs $f(x, y)$.

Thus any Boolean function can be computed with 1 bit communication.

4 A perfect magic trick

(a)

Each player fixes deterministic answers:

$$a_{\top}, a_{\perp}, \quad b_{\top}, b_{\perp}, \quad c_{\top}, c_{\perp}.$$

Write parity constraints:

$$a_{\top} + b_{\top} + c_{\top} = 0 \pmod{2},$$

$$a_{\top} + b_{\perp} + c_{\perp} = 1,$$

$$a_{\perp} + b_{\top} + c_{\perp} = 1,$$

$$a_{\perp} + b_{\perp} + c_{\top} = 1.$$

Adding all four equations gives contradiction mod 2. Hence at least one condition fails.

Maximum success probability $3/4$.

(b)

Given state:

$$\frac{1}{2}(|000\rangle - |011\rangle - |101\rangle - |110\rangle).$$

Brute Force Approach: In $\top\top\top$ case, measurement basis $\{|0\rangle, |1\rangle\}, \{|0\rangle, |1\rangle\}, \{|0\rangle, |1\rangle\}$. So, get measurement results either of the 4 000, 011, 101 or 110. Check all of them satisfy even number of 1s.

In the $\top \perp \perp$ case, measurement basis $\{|0\rangle, |1\rangle\}, \{|+\rangle, |-\rangle\}, \{|+\rangle, |-\rangle\}$. Suppose, Alice has measured in $|0\rangle$. Bob and Charlie's state collapses in $\frac{1}{\sqrt{2}}[|00\rangle - |11\rangle]$. So, Bob and Charlie see any 1 of the 4 outcomes $++$, $+-$, $-+$, $--$ on their joint state. Probability that they see $++$ is $|\langle ++, \frac{1}{\sqrt{2}}[|00\rangle - |11\rangle]\rangle|^2 = \frac{1}{2}|\langle ++, 00\rangle - \langle ++, 11\rangle|^2 = 0$ Check similarly measurement probability of $--$ is 0 and that of $+-$, $-+$ are $1/2$ - $1/2$ each. Thus we get final outcomes either $0(\text{Alice}) + (\text{Bob}) - (\text{Charlie})$ or $0(\text{Alice}) - (\text{Bob}) + (\text{Charlie})$. Note, $+/-$ outcomes are mapped to 0/1. Hence we get odd number of 1s.

Check the same thing happens if Alice measures 1.

Check everytime we get odd number of 1s of $\perp \top \perp$ and $\perp \perp \top$.

Observable Trick discussed in tutorials: Observable corresponding to $|0\rangle, |1\rangle$ basis is Z and $|0\rangle, |1\rangle$ basis is X if the outcomes 0/1 are mapped to ± 1 . Product of $(-1)^{\text{outcomes}}$ is equal to eigen value of the Observable (which eigen state the measurement collapses in). We are done by noting that $Z \otimes Z \otimes Z |\psi\rangle = 1 \cdot |\psi\rangle$ and $Z \otimes X \otimes X |\psi\rangle = (-1) \cdot |\psi\rangle, X \otimes Z \otimes X |\psi\rangle = (-1) \cdot |\psi\rangle, X \otimes X \otimes Z |\psi\rangle = (-1) \cdot |\psi\rangle$,

5 Hardy's Paradox

(a)

Expand state explicitly.

One checks:

- TT: amplitude of $|11\rangle$ nonzero $\rightarrow$ possible.
- T,H: amplitude of $|10\rangle$ is 0.
- H,T: amplitude of $|01\rangle$ is 0.
- H,H: amplitude of $|11\rangle$ is 0.

All follow from direct expansion.

(b)

Lucien assumes pre-existing measurement outcomes.

But quantum measurement outcomes depend on measurement choice.

The reasoning assumes non-contextual hidden variables.

Hardy's paradox shows such assignment leads to contradiction.

Thus Lucien's reasoning is invalid.

6 Multiplicative generators modulo a prime

Will be discussed in a video tutorial soon. Questions won't be asked on this topic in Quiz 1.